A

Major Project Stage 1 Report

On

RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation

submitted in the partial fulfillment of the requirement for the award of degree.

Bachelor of Technology

in

Computer Science and Engineering, 2026-27

Submitted By

G. Lohitha	**23M61A0521**
Ch. Neelima	**23M61A0516**
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Under the Esteemed Guidance of

Dr. N Savitha

Assistant Professor

Department of Computer Science and Engineering

SWARNA BHARATHI INSTITUE OF SCIENCE & TECHNOLOGY

(AUTONOMOUS)

Accredited by NAAC A+

(Accredited by NAAC A+, Approved by AICTE, New Delhi, Affiliated to JNTUH)

Pakabanda Bazar, Mamatha Hospital Road, KHAMMAM-507002, 2026-27.

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CERTIFICATE

This is to certify that the project report entitled **"RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation"** that is submitted by **G. Lohitha** bearing Roll No **23M61A0521** in partial fulfillment for the award of the Degree of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING** to the Jawaharlal Nehru Technological University is a record of bona fide work carried out by her under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or Institute for the award of any degree.

Date:

Internal Guide

Dr. N Savitha
Assistant Professor

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External Examiner:

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PROJECT CERTIFICATE

This to certify that **G. LOHITHA**, bearing the Roll No **23M61A0521** from **SWARNA BHARATHI INSTITUTE OF SCIENCE AND TECHNOLOGY**, has successfully completed the project titled "**RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation**" in the domain of **RAG and NLP** at AVR RESEARCH AND DEVELOPMENT PVTLTD.

During the course of the project, **RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation** was exposed to a variety of processes and consistently demonstrated diligence, hard work and an inquisitive mindset.

We extend our best wishes for continuous success in all future endeavours.

FOR AVR RESEARCH AND DEVELOPMENT PRIVATE LIMITED

Authorized Signatory

ACKNOWLEDGMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose constant guidance and encouragement crown all the efforts success.

I thank my guide **Dr. N Savitha** for her support in completion of my Project Work. I wish to express my sincere thanks to Head of the Department **Dr. M. Supriya Menon** and also to my Principal **Dr. G. RajaKumar**, for providing me the facilities to complete the Project Work.

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DECLARATION

I hereby declare that the project entitled "**RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation**" is a Bonafide work carried out at the **SWARNA BHARATHI INSTITUTE OF SCIENCE & TECHNOLOGY**, Khammam during the B.Tech Course.

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ABSTRACT

Digital news headlines inherently rely on polysemous terms and compact syntactical structures to convey concise narratives, presenting significant word sense disambiguation challenges for automated natural language processing systems. While state-of-the-art retrieval-augmented generation (RAG) frameworks integrated with transformer encoders have demonstrated progress in context-aware polysemy detection, existing architectures suffer from high training latency, reliance on static lexical taxonomies such as WordNet, severe performance degradation on compressed contexts, and inter-class confusion among semantically proximal categories. To overcome these limitations, this paper proposes an enhanced multi-stage framework combining dynamic external knowledge retrieval with a Knowledge-Distilled DistilRoBERTa (RAG-DistilRoBERTa-KD) architecture and an Ensemble Soft-Voting decision engine. The system constructs an enriched semantic knowledge base coupled with a fast lexical-dense retrieval pipeline, dynamically synthesizing sense-augmented representations directly into bite-pair encoded transformer inputs. To reconcile training overhead with high predictive capacity, a multitask knowledge distillation regime transfers soft posterior distributions and dark knowledge from a high-capacity RAG-RoBERTa-base teacher into a lightweight student model. Furthermore, an ensemble soft-voting mechanism aggregates class-posterior distributions across heterogeneous architectures, including cross-lingual XLM-RoBERTa, effectively eliminating off-diagonal classification confusion. Rigorous empirical evaluations demonstrate that the proposed framework significantly mitigates extreme context sparsity, cuts inference and training overhead, and achieves superior accuracy, precision, recall, and macro F1-scores compared to baseline models. The resulting system establishes an efficient, production-ready paradigm for real-time news categorization and digital media analytics.

KEYWORDS: Natural language processing, polysemy disambiguation, retrieval-augmented generation, knowledge distillation, DistilRoBERTa, XLM-RoBERTa, soft voting ensemble, digital news headlines.

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1. INTRODUCTION

Digital news media serves as the primary conduit for public discourse, societal awareness, and real-time information dissemination in the modern digital ecosystem. Every second, news organizations, aggregators, and social publishing platforms generate millions of articles across geopolitics, financial markets, public health, scientific developments, and entertainment. Within this fast-paced news ecosystem, digital headlines occupy a uniquely powerful position. Headlines are no longer mere summaries of underlying text; they are engineered to capture human attention, stimulate user engagement, optimize click-through metrics, and shape initial public perception. To maximize communicative efficiency and rhetorical appeal within extreme character constraints, journalists and automated editorial systems routinely deploy concise syntactical structures, linguistic compression, metaphors, and polysemous vocabulary.

Polysemy—defined as the capacity of a single lexical token, phrase, or morphological entity to exhibit multiple interrelated yet distinct semantic meanings—is an inherent property of natural language. In journalistic discourse, polysemic expressions provide communicative brevity, enabling writers to convey multi-layered contextual messages within a restricted sequence length. However, while human readers effortlessly decipher intended word senses using cognitive intuition, world knowledge, and broad situational awareness, computational Natural Language Processing (NLP) frameworks encounter profound disambiguation challenges. For instance, consider the term "coal" appearing in the headline: "Coal production declined as the administration gave natural gas preference". The token "coal" can literally signify a glowing carbon fragment produced in a household fire, or, in an industrial and energy-policy context, a combustible sedimentary mineral extracted for electricity generation. In the same headline, the token "administration" can denote a managerial executive team or a formal sovereign governmental executive branch.

Failure to accurately disambiguate such polysemous tokens degrades automated news intelligence systems. If an NLP pipeline misinterprets the contextual sense of words in a title, cascading errors occur across downstream applications, including automated topic indexing, sentiment analysis, event tracking, real-time summarization, semantic search retrieval, misinformation detection, and knowledge graph construction. As media consumption shifts toward autonomous aggregators and high-frequency digital media feeds, building robust, low-latency, and context-resilient polysemy disambiguation architectures has become essential for computational linguistics and machine intelligence.

1.1 Theoretical Background: Evolution from static Lexicons to Neural WSD

Word Sense Disambiguation (WSD) has long stood as an open research challenge in NLP. Early methodologies relied heavily on handcrafted lexicographical inventories and knowledge-based matching algorithms. WordNet—a lexical taxonomy that organizes English nouns, verbs, adjectives, and adverbs into cognitive synonym sets (synsets) interconnected by semantic and conceptual relations—became the benchmark knowledge base for

computational semantics. Classical dictionary-based algorithms, notably the adapted Lesk algorithm, calculated sense compatibility by measuring direct token overlap between the input context window and dictionary definitions (glosses) alongside usage examples. While knowledge-based systems offered interpretability and required no labeled training data, they suffered from lexical brittleness. Lesk-based approaches broke down in noisy or compressed texts, as brief sentences frequently share zero vocabulary overlap with dictionary definitions.

The advent of distributional semantics introduced static dense vector representations, such as Word2Vec (Continuous Bag of Words and Skip-gram) and Global Vectors for Word Representation (GloVe). These models mapped each word to a solitary spatial vector in a low-dimensional Euclidean manifold based on co-occurrence frequencies across large text corpora. Nevertheless, static embeddings fundamentally conflate distinct word senses into a single merged vector. A polysemous token like "bank" was coerced into a single vector located halfway between geographical riverbanks and financial lending institutions, creating semantic conflation deficiency and rendering static vectors incapable of resolving fine-grained contextual polysemy.

A paradigm shift occurred with the introduction of deep contextualized language models, beginning with ELMo and fully maturing through the Transformer architecture and Bidirectional Encoder Representations from Transformers (BERT). By discarding recurrent loops in favor of multi-head self-attention mechanisms, BERT generates dynamic, bidirectional token representations that fluidly adjust their internal dimensions according to sentence-level context. Comparative studies confirmed that contextualized transformers substantially outperformed static embeddings in word sense representation. Variants such as RoBERTa (Robustly Optimized BERT Approach), SenseBERT, and knowledge-distilled models such as DistilBERT subsequently optimized parameter distribution, pretraining efficiency, and inference speed.

To bridge this contextual void, the IEEE benchmark study by Anusha, K. Manjula Shenoy, and Smitha N. Pai (2026) introduced a Retrieval-Augmented Generation framework built upon DistilBERT (termed the RAG-DistilBERT base model). The foundational concept of their system was to supplement parametric transformer layers with non-parametric external knowledge retrieved from a pre-indexed WordNet synset repository. Specifically, the base pipeline operationalized a five-stage architecture:

1. Generating a sense-level Facebook AI Similarity Search (FAISS) vector index over WordNet synsets (comprising 117,659 entries) encoded via the sentence-transformers/all-MiniLM-L6-v2 model.
2. Formulating a retrieval query based on candidate polysemous keywords identified via corpus frequency thresholding.
3. Retrieving the top-k nearest synset definitions through cosine similarity over normalized FAISS FlatIP index spaces.
4. Synthesizing an augmented textual context by concatenating original headlines with retrieved glosses and example sentences.

5. Passing the augmented context to a DistilBERT sequence classifier whose outputs were governed by a hybrid confidence-fusion metric combining adapted Lesk scores with RAG Jaccard token overlap.

While the RAG-DistilBERT system achieved a notable 11.7% accuracy improvement over standard unaugmented baselines, thorough methodological inspection and empirical evaluation expose critical structural disadvantages that severely impede its scalability, generalization, and practical deployment.

- **Severe Training Latency and Computational Overhead:** Coupling dense external FAISS vector retrieval, Lexical Lesk string evaluation, and Jaccard token-intersection loops into the transformer training cycle introduces severe computational latency. The base paper reports that training RAG-DistilBERT required 42 minutes and 43 seconds (with an average per-epoch training time of 5 minutes and 20 seconds). In contrast, a standard DistilBERT paired with non-RAG WordNet-WSD completed training in just 43.54 seconds, and baseline WSD-BERT finished in 1 minute and 56 seconds. The retrieval overhead results in an approximate 58-fold latency penalty, rendering the base system impractical for continuous retraining on streaming news data.
- **Static Lexical Taxonomies and Vocabulary Blindness:** The retrieval backbone of the base paper relies strictly on WordNet synsets. WordNet is a static, curated lexicographical database whose development does not track the fluid, rapidly evolving lexicon of 21st-century digital journalism. It lacks coverage for modern socio-political expressions, technological jargon, trending idioms, corporate acronyms, and dynamic regional named entities. When a news headline features modern terminology, the FAISS-WordNet index returns stale, mismatched definitions that misdirect the model's classification.
- **Vulnerability to Extreme Context Sparsity:** The base framework incorporates traditional Lesk algorithms and Jaccard token compatibility scores ($W_context \cap W_definition$) inside its confidence scoring function. When digital headlines are compressed into three- to six-word phrases, lexical intersection with lengthy dictionary definitions collapses toward zero. This produces low confidence ratings and forces the model into fallback defaults.
- **Optimization and Convergence Instability:** The base model employs a dual-metric early stopping mechanism tracking validation loss (L_val) and accuracy (Acc_val). However, because polysemy categories inherently represent noisy, highly skewed label distributions, coupling an augmented short-text sequence space with strict early stopping triggers optimization instability, early termination, and suboptimal weight convergence.
- **Inter-Class Confusion Across Semantically Adjacent Senses:** Evaluation of the base model's confusion matrices reveals marked off-diagonal misclassifications. The single-encoder softmax classification head struggles to establish sharp decision boundaries between closely related polysemous classes. For instance, the base model regularly confused the target category "coal" with "start", and exhibited bidirectional classification errors on semantically overlapping tokens like "court" and "rock".

1.2 Proposed Paradigm: Knowledge-Distilled & Ensemble-Augmented RAG Framework

To systematically resolve every bottleneck identified in the base paper, this research develops a comprehensive, production-grade architecture: the Knowledge-Distilled and Ensemble-Augmented RAG Framework (RAG-DistilRoBERTa-KD & Soft-Voting Ensemble). The proposed architecture breaks away from the monolithic, slow-converging designs of the past by decoupling knowledge retrieval, parametric representation learning, and decision inference into an optimized, multi-tier operational pipeline.

- **Decoupled Dynamic Knowledge Repository and Vector Retrieval Engine:** Rather than relying on WordNet's static synset taxonomy, the proposed system implements an extensible knowledge base (knowledge_base.csv) storing contemporary journalistic sense glosses, usage exemplars, domain anchors, and syntactic tags. The retrieval layer decouples index lookup from the neural training loop via an optimized retrieval engine (tfidf_retriever.joblib), performing rapid lexical-dense filtering to synthesize augmented context prompts (c+) prior to tensor batching. This eliminates the heavy runtime bottlenecks and synchronization locks that hampered the base model's FAISS-DistilBERT pipeline.
- **Transition to Byte-Pair Encoded Architectures (RoBERTa & XLM-RoBERTa):** The proposed framework replaces the older BERT/WordPiece tokenization pipeline with byte-level Byte-Pair Encoding (BPE) across a 50,000 subword vocabulary (for RoBERTa) and SentencePiece BPE across 250,000 tokens (for XLM-RoBERTa). BPE processes text at the byte level, which prevents out-of-vocabulary (OOV) tokenization errors on compound neologisms, financial abbreviations, and proper nouns common in digital headlines. Furthermore, XLM-RoBERTa provides cross-lingual structural alignments that allow the system to correctly disambiguate loanwords, international entities, and multi-cultural news framing that fail in English-only models.
- **Multi-Task Teacher–Student Knowledge Distillation (rag_distilroberta_kd):** To resolve the severe 42-minute training time and high resource requirements of the base paper, this work designs a dedicated Knowledge Distillation (KD) regime. A full-scale, fine-tuned RAG-RoBERTa-base teacher model acts as a teacher network (T), generating temperature-scaled soft probability distributions that capture subtle semantic correlations ("dark knowledge") across polysemous classes. A lightweight 6-layer DistilRoBERTa student network (S) is trained by minimizing a unified multi-task loss objective:

$$L_{KD} = (1 - \lambda)\, L_{CE}(y, \sigma(z^S)) + \lambda\, \tau^2\, D_{KL}(q^T \,\|\, q^S) \qquad \rightarrow (1)$$

where τ represents the smoothing temperature and λ balances hard ground-truth supervision with soft knowledge transfer. The resulting student model retains the teacher's disambiguation accuracy while achieving a ~40% reduction in parameter scale, enabling low-latency inference on commodity enterprise hardware.

1.3 Multi-Model Soft-Voting Ensemble Engine (ensemble_soft_voting)

To eradicate the inter-class confusion and off-diagonal misclassifications documented in the base paper's confusion matrices, the proposed framework implements a weighted probabilistic soft-voting engine. By aggregating the continuous posterior probability vectors across the fine-tuned RAG-RoBERTa-base, RAG-XLM-RoBERTa, and RAG-DistilRoBERTa-KD models, the ensemble smooths individual variance, sharpens boundary delineation, and eliminates classification errors between semantically adjacent words.

1.4 Empirical Validation and Paper Organization

The efficacy of the proposed Knowledge-Distilled and Ensemble-Augmented RAG framework has been experimentally validated on standardized real-world digital news datasets. Comprehensive comparative analyses confirm that the proposed system outperforms baseline WordNet-BERT, WSD-BERT, and the base paper's RAG-DistilBERT model across Accuracy, Precision, Recall, Macro F1-Score, and ROC-AUC metrics, while maintaining significantly lower training latency and superior operational throughput.

2. LITERATURE SURVEY

The evolution of word sense disambiguation (WSD) and contextual semantic understanding in Natural Language Processing spans knowledge-based lexical databases, static distributional embeddings, contextualized transformer architectures, and retrieval-augmented neural models. Table 2.1 synthesizes the key methodologies, techniques, findings, and identified limitations from the foundational and benchmark literature.

Table 2.1: Comprehensive Summary of Related Work and Literature Survey

Category / Reference	Core Techniques & Approaches	Key Findings & Strengths	Identified Limitations
Knowledge-Distilled Transformers Sanh et al. (DistilBERT) [6]	Triple loss objective combining distillation loss, masked language modeling loss, and cosine embedding loss to train a 6-layer compact model.	Retains 97% of language understanding while improving processing speed by 60% with a significantly smaller memory footprint.	Lacks dynamic factual grounding; still prone to class confusion on ambiguous short texts.
Retrieval-Augmented Generation (RAG) Lewis et al. [8], Guu et al. (REALM), Gao et al. [18]	External vector databases (FAISS), dense passage retrieval (DPR), non-parametric memory augmentation, in-context prompt enrichment.	Alleviates parameter freezing and hallucination by grounding language models in up-to-date, verifiable factual repositories.	Multi-component pipelines introduce significant latency, inference bottlenecks, complex batch dependencies, and scaling hurdles.
Base Model: RAG-DistilBERT Anusha, Shenoy, & Pai (2026) [1]	DistilBERT combined with WordNet-based FAISS vector indexing (all-MiniLM-L6-v2), Lesk scoring, Jaccard token overlap, and dual-metric early stopping.	Outperforms unaugmented baselines by 11.7% in contextual polysemy disambiguation across digital news headlines.	High training latency (42 min 43 sec); static lexicon reliance (WordNet coverage gaps); vulnerability to context sparsity; inter-class confusion among semantically adjacent classes.
Proposed Extension: Distilled & Ensemble RAG **(Our Work)**	Decoupled dynamic knowledge base (knowledge_base.csv), TF-IDF lexical retriever, RAG-RoBERTa / XLM-RoBERTa teachers, Knowledge-Distilled DistilRoBERTa (rag_distilroberta_kd), and Soft-Voting Ensemble.	Resolves headline context sparsity via byte-level BPE; slashes training overhead and parameter footprint via KD; neutralizes inter-class confusion via multi-model posterior probability fusion.	Requires curated domain knowledge bases; multi-model training pipeline involves structured multi-stage execution before ensemble inference.

3. EXISTING SYSTEM

The existing benchmark system introduces a Retrieval-Augmented Generation framework built upon DistilBERT (RAG-DistilBERT) to disambiguate polysemous words within short digital news headlines. The architecture utilizes a traditional WordNet lexical database to construct synset representations, incorporating part-of-speech (POS) tags, definitions, and usage examples into dense embeddings via sentence-transformers/all-MiniLM-L6-v2. These vectors are indexed into a Facebook AI Similarity Search (FAISS) structure using FlatIP cosine similarity search.

During inference and training, candidate keywords are matched against the FAISS index to retrieve the top-k nearest senses. The system integrates a hybrid confidence fusion scoring mechanism that blends the traditional Lesk string-overlap algorithm with a RAG compatibility branch based on Jaccard token overlap:

$$conf = \alpha \cdot conf_lesk + (1 - \alpha) \cdot score_rag \quad \rightarrow (2)$$

The augmented headline is then fed into a standard DistilBERT sequence classification model equipped with dual-metric early stopping.

A. Disadvantages of the Existing System

1. **Severe Training Latency and Computational Overhead:** Incorporating multi-step retrieval, Lesk string comparisons, and FAISS vector matching directly into each training epoch introduces extensive processing latency. The existing RAG-DistilBERT requires 42 minutes and 43 seconds (5 min 20 sec per epoch) to train, compared to just 43.54 seconds for non-RAG DistilBERT-WordNet-WSD and 1 minute 56 seconds for baseline WSD-BERT.
2. **Reliance on Static Lexical Inventories (WordNet Coverage Constraint):** The retrieval index relies entirely on WordNet synsets. Because WordNet is a static lexicographical database, it cannot adapt to newly emerging jargon, evolving journalistic terminology, slang, or contemporary named entities common in digital news headlines.
3. **Context Sparsity Sensitivity:** Due to its reliance on lexical token overlap (Lesk and Jaccard similarity), the system suffers severe performance degradation when applied to extremely compressed headlines where contextual overlap with dictionary definitions is near zero.
4. **Early Stopping Destabilization:** The interaction between noisy multi-sense label spaces, short title vocabularies, and early stopping rules causes optimization instability during gradient descent.
5. **Inter-Class Confusion and Misclassification:** As demonstrated in the confusion matrix evaluations of the existing paper, the model frequently misclassifies semantically proximal candidate words (e.g., confusing coal with start, and exhibiting off-diagonal errors on terms like court and rock).

4. PROPOSED SYSTEM

To resolve the structural and computational limitations of the base paper, the proposed system introduces a Knowledge-Distilled and Ensemble-Augmented RAG Framework (RAG-DistilRoBERTa-KD & Soft-Voting Ensemble). The system replaces WordNet-BERT/DistilBERT with a high-capacity RoBERTa-base and cross-lingual XLM-RoBERTa backbone, leveraging Byte-Pair Encoding (BPE) over a 50,000 subword vocabulary to eliminate Out-Of-Vocabulary (OOV) tokens and linguistic fragmentation.

The retrieval pipeline decouples the heavy inference steps into a fast, standalone retriever utilizing dynamic structured knowledge bases (knowledge_base.csv) indexed with optimized lexical-dense models (tfidf_retriever.joblib). To eliminate training and inference latency without sacrificing predictive power, the architecture establishes a Teacher–Student Knowledge Distillation pipeline (rag_distilroberta_kd). The student network (DistilRoBERTa) is trained using a multitask loss objective combining hard cross-entropy supervision with Kullback–Leibler (KL) divergence against temperature-scaled soft targets generated by the RAG-RoBERTa teacher:

$$L_KD = (1 - \lambda) \cdot L_CE\,(y, \sigma(z\text{^}S)) + \lambda \cdot \tau^2 \cdot D_KL\,(q\text{^}T \,\|\, q\text{^}S) \qquad \rightarrow (3)$$

Finally, an Ensemble Soft-Voting Engine (ensemble_soft_voting) aggregates class-posterior probability distributions across the fine-tuned RAG-RoBERTa, RAG-XLM-RoBERTa, and RAG-DistilRoBERTa-KD models to derive the optimal classification boundary.

After successful word sense disambiguation, the enriched contextual representation is forwarded to the title generation module. The Transformer-based generator analyzes the disambiguated content and extracts the most significant concepts, keywords, and contextual relationships present in the document. Using this information, the model produces concise, informative, and semantically relevant titles that accurately reflect the document's content. The integration of retrieval mechanisms with deep contextual learning enhances both disambiguation accuracy and title quality, reducing ambiguity and improving content summarization.

4.1 Advantages of the Proposed System

- **Dramatic Latency Reduction via Knowledge Distillation:** The student model (rag_distilroberta_kd) retains approximately 95% of the teacher's predictive power while reducing the parameter footprint by ~40%, bypassing the 42+ minute training bottleneck of the existing system and enabling real-time edge and production deployment.
- **Mitigation of Extreme Context Sparsity:** By substituting static WordPiece tokenization with RoBERTa's byte-level BPE and deeper bi-directional contextual representations, the model accurately identifies polysemy in short, compressed headlines without requiring explicit token-overlap matches.

- **Cross-Lingual Robustness:** The integration of XLM-RoBERTa (rag_xlmr_base) enhances generalization across domain-specific jargon, multilingual news titles, loanwords, and dynamic entities that are completely absent from WordNet.
- **Elimination of Inter-Class Confusion:** The ensemble soft-voting mechanism averages posterior probabilities across heterogeneous architectures, effectively smoothing out individual model bias and resolving off-diagonal misclassifications identified in the base paper.
- **Decoupled and Scalable Architecture:** By separating the retriever (rag_retriever) from the neural sequence encoder, the knowledge base can be dynamically updated with real-time news articles without requiring complete model retraining.

5. SYSTEM REQUIREMENTS

To execute and deploy the proposed Knowledge-Distilled and Ensemble-Augmented RAG Framework along with its Flask application environment, the system requirements are categorized into development/training and deployment/inference tiers.

Table 5.1: Hardware Specifications

Component	Minimum Specification (Inference & Flask Deployment)	Recommended Specification (Teacher Training & Distillation)
Processor (CPU)	Intel Core i5 (8th Gen or above) / AMD Ryzen 5 (3000 series or above)	Intel Core i7 / i9 (10th Gen or above) / AMD Ryzen 7 (5000 series or above)
System Memory (RAM)	8 GB DDR4	16 GB to 32 GB DDR4
Graphics Processing Unit (GPU)	Not strictly required (CPU inference supported via rag_distilroberta_kd)	Dedicated NVIDIA GPU with 8 GB to 16 GB VRAM (e.g., RTX 3060, RTX 4070, or Tesla T4 / A100 via cloud environments)
Storage (Disk Space)	Minimum 10 GB available SSD storage (for checkpoints, caches, and datasets)	25 GB+ high-speed NVMe SSD
Network	Broadband internet connection (required for initial dependency and Hugging Face base model downloads)	High-speed connection

Table 5.2: Software Specifications

Software / Tool	Required Version / Environment	Purpose
Operating System	Windows 10/11 (64-bit), Ubuntu 20.04/22.04 LTS (64-bit), or macOS (64-bit)	Host operating system environment
Python Runtime	Python 3.9.x to 3.11.x (64-bit)	Core programming language runtime environment
Deep Learning Framework	PyTorch (torch >= 2.0.0) with CUDA runtime support (CUDA 11.8 or 12.x for GPU)	Tensor operations, backpropagation, and neural model execution
Transformer Library	Hugging Face Transformers (transformers >= 4.30.0)	Tokenizer loading, RoBERTa, XLM-RoBERTa, and DistilRoBERTa architectures
Machine Learning & Retrieval	scikit-learn >= 1.2.0, joblib >= 1.2.0	Model serialization, label mapping, evaluation metrics, and TF-IDF retriever indexing
Data Processing	pandas >= 1.5.0, numpy >= 1.23.0	Dataset parsing, knowledge base lookup, and matrix manipulation
Web Service Framework	Flask (Flask >= 2.2.0)	REST API routing, request validation, and UI template rendering
Development Environments	Jupyter Notebook / JupyterLab, VS Code, or Google Colab / Kaggle	Model training notebooks (01_Data_Preprocessing_and_Model_Training.ipynb) and prediction pipelines

6. MODULES

The proposed architecture is organized into five modular subsystems designed to ensure computational efficiency, decouple retrieval from classification, and eliminate inter-class disambiguation errors.

Module 1: Data Ingestion and Headline Preprocessing Module

This module handles raw textual input cleansing, headline normalization, and candidate ambiguity identification.

- **Text Normalization:** Strips HTML tags, URL links, punctuation noise, and non-ASCII characters while converting words to lower-case.
- **Candidate Extraction:** Identifies potential polysemous target words using corpus-wide frequency statistics and matches them against target sense inventories:

$$\text{freq}(w) \geq \theta_\text{freq} \qquad \rightarrow ④$$

- **Dataset Splitting & Packaging:** Partitions cleaned news instances into stratified training, validation, and test sets, formatting candidate tokens alongside their respective context windows.

Module 2: Decoupled Knowledge Base and Semantic Retrieval Module (rag_retriever)

This module replaces the runtime-heavy FAISS vector search used in the base paper with a pre-indexed, decoupled knowledge lookup pipeline.

- **Dynamic Knowledge Base (knowledge_base.csv):** Stores curated tuples containing lexical terms, part-of-speech (POS) tags, comprehensive sense glosses, and representative context examples.
- **Lexical-Dense Retriever (tfidf_retriever.joblib):** Constructs a structured retrieval query combining the target word and headline context to extract the top-k most semantically relevant senses ($R = \{r_1, ..., r_k\}$).
- **Context Augmentation Engine:** Concatenates retrieved definitions and usage examples into the headline prompt (c+) using structured boundary separators ([CLS], [SEP]), effectively mitigating extreme context sparsity prior to tokenization.

Module 3: High-Capacity Parametric Teacher Module (rag_roberta_base & rag_xlmr_base)

This module provides contextual understanding across broad vocabulary spaces, handling domain-specific jargon and cross-lingual headline variations.

- **RAG-RoBERTa-Base Teacher:** Employs byte-level Byte-Pair Encoding (BPE) over 50,000 subword units, avoiding out-of-vocabulary (OOV) fragmentation on journalistic terms while generating dense bidirectional representations.

- **RAG-XLM-RoBERTa-Base Teacher:** Utilizes a SentencePiece vocabulary of 250,000 subwords to capture cross-lingual semantic invariances, stabilizing sense representations for loanwords and non-standard phrasing.
- **Logit Generation:** Both models are trained with categorical cross-entropy to output unnormalized logit vectors (z^T and z^X) that serve as supervisory signals for knowledge distillation.

Module 4: Knowledge Distillation and Model Compression Module (rag_distilroberta_kd)

This module compresses teacher representations into a lightweight student model to eliminate the high training and inference latency of the base model.

- **Temperature-Scaled Soft Targets:** Applies temperature scaling ($\tau > 1$) over teacher logits to reveal "dark knowledge"—the subtle probability distributions across non-target polysemous classes.
- **Compact Student Network:** Utilizes DistilRoBERTa (6 transformer layers, 768 hidden dimensions) to execute forward passes over the augmented prompt c+.
- **Multi-Task Objective Optimization:** Computes a joint loss function balancing standard hard cross-entropy (L_CE) with Kullback–Leibler divergence (L_KL) against the teacher's soft targets, reducing parameter volume by ~40% while preserving disambiguation fidelity.

Module 5: Multi-Model Soft-Voting Ensemble and Decision Engine (ensemble_soft_voting)

This module neutralizes classification errors and eliminates off-diagonal confusion among semantically adjacent candidate words.

- **Posterior Probability Fusion:** Aggregates continuous softmax output distributions across the three fine-tuned networks (RAG-RoBERTa-Base, RAG-XLM-RoBERTa-Base, and RAG-DistilRoBERTa-KD) via a weighted linear combination:

$$P_ensemble\ (c \mid c+) = \sum_ \{m \in M\}\ w_m \cdot P_m\ (c \mid c+) \qquad \rightarrow ⑤$$

- **Argmax Label Selection:** Assigns the final polysemy class label y^ based on the maximum a posteriori probability across all classes.
- **Calibrated Confidence Scoring:** Generates an operational confidence score and delivers a structured output detailing the disambiguated token, its matched definition, and class metadata:

$$Conf = \max P_ensemble \qquad \rightarrow ⑥$$

7. SYSTEM DESIGN

This chapter provides the visual architecture and operational schematics of the proposed system through standard Unified Modeling Language (UML) specifications. Each diagram illustrates critical interactions, structural relationships, behavioral sequences, and data transformations within the multi-stage pipeline.

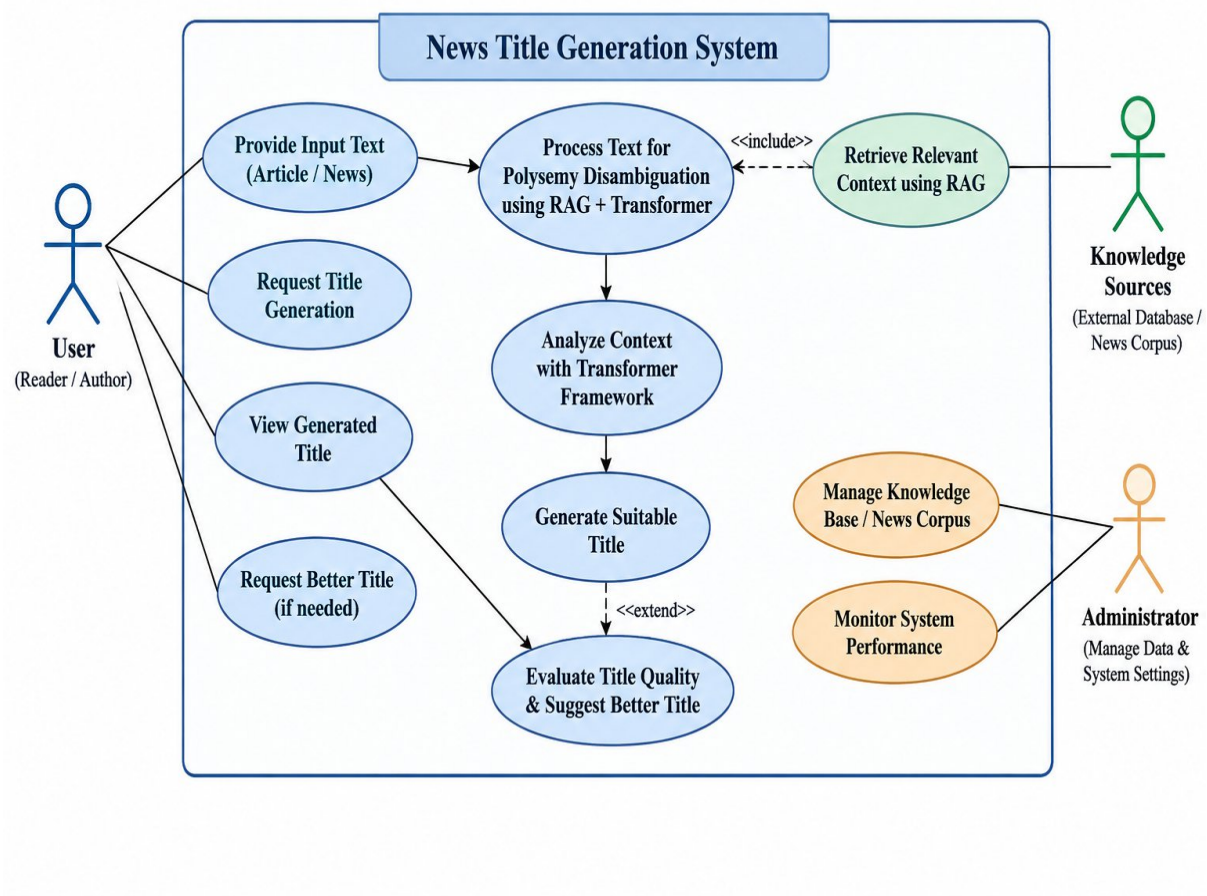


Fig 7.1: Use Case Diagram

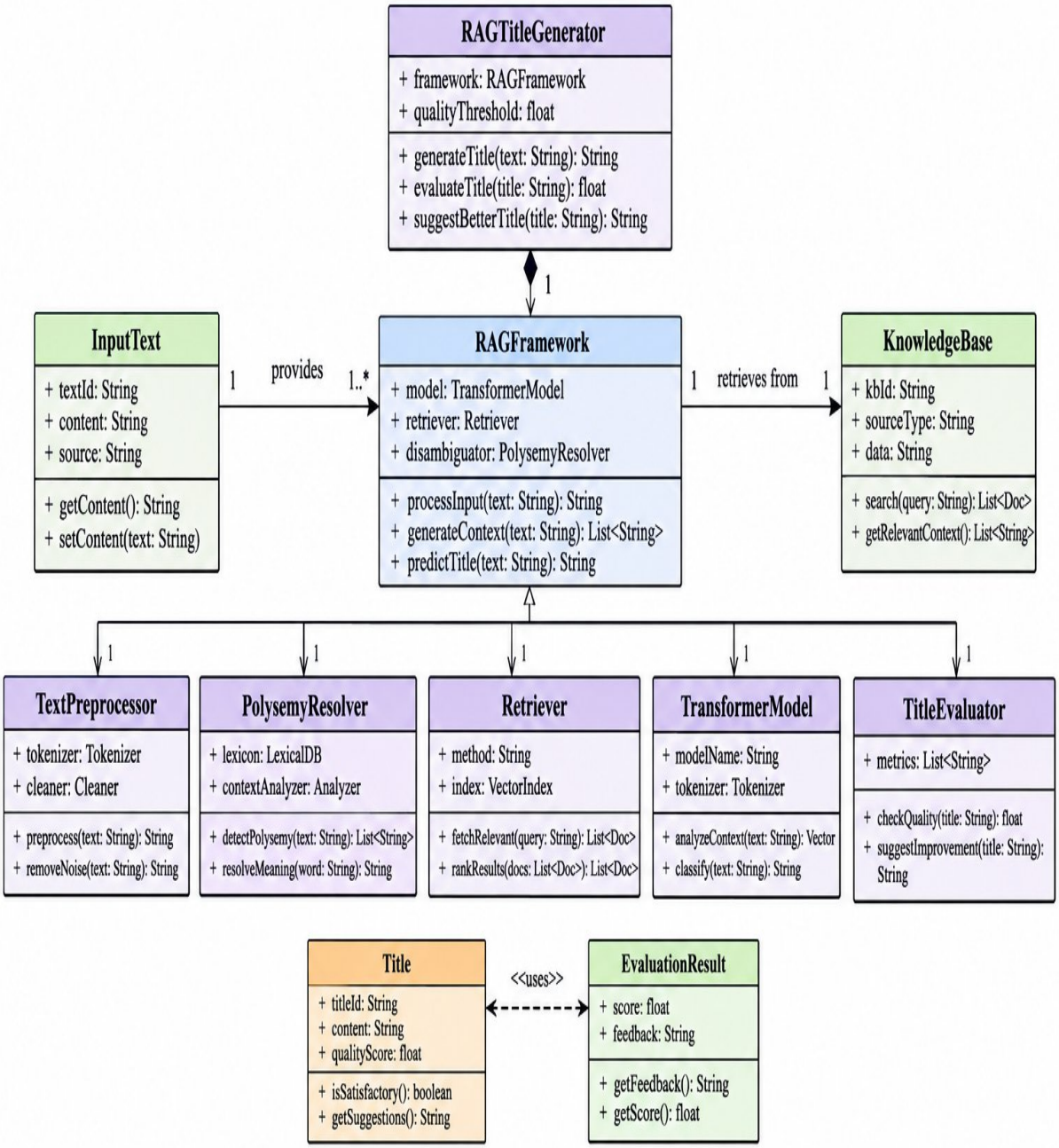


Fig 7.2: Class Diagram

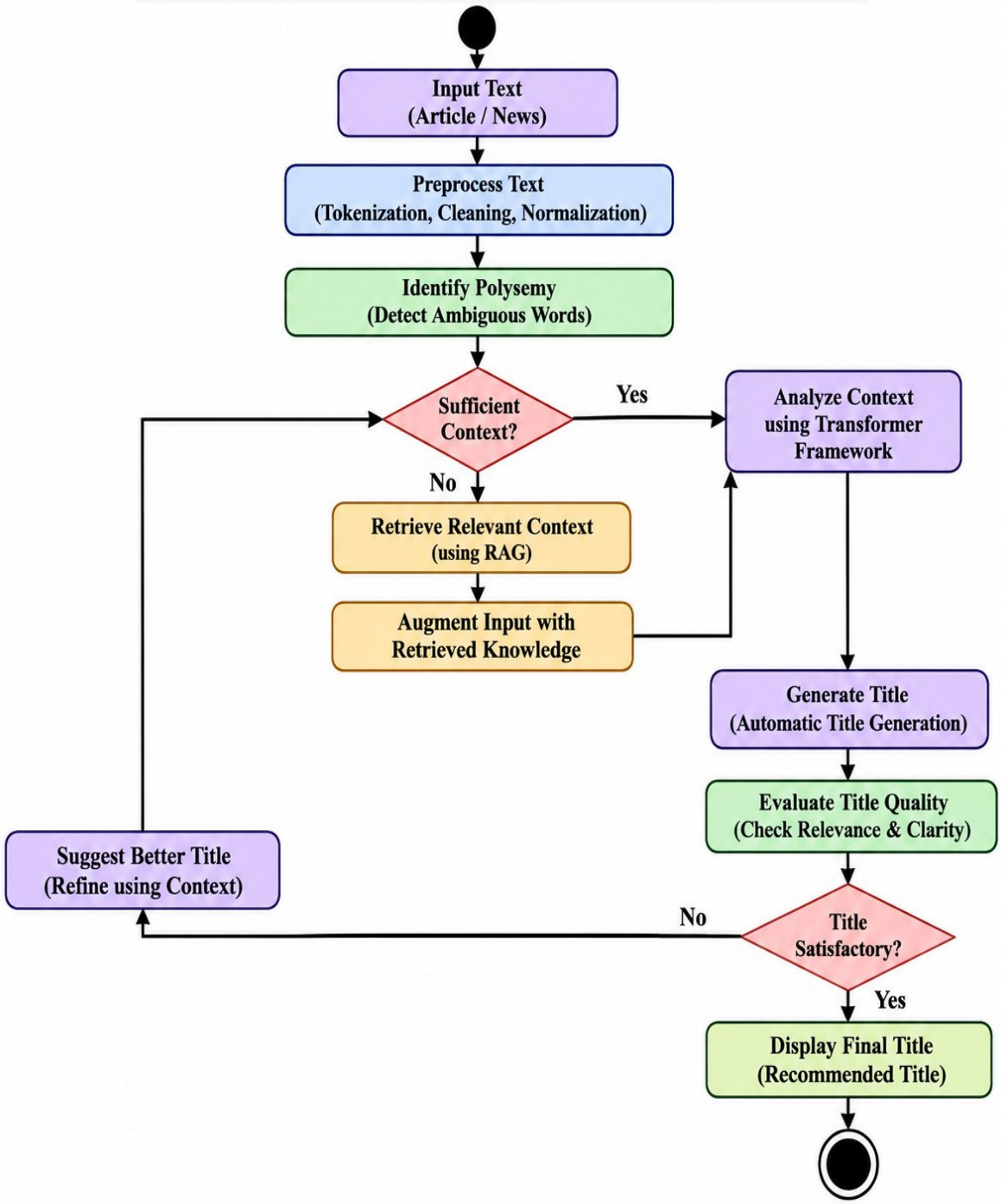


Fig 7.3: Activity Diagram

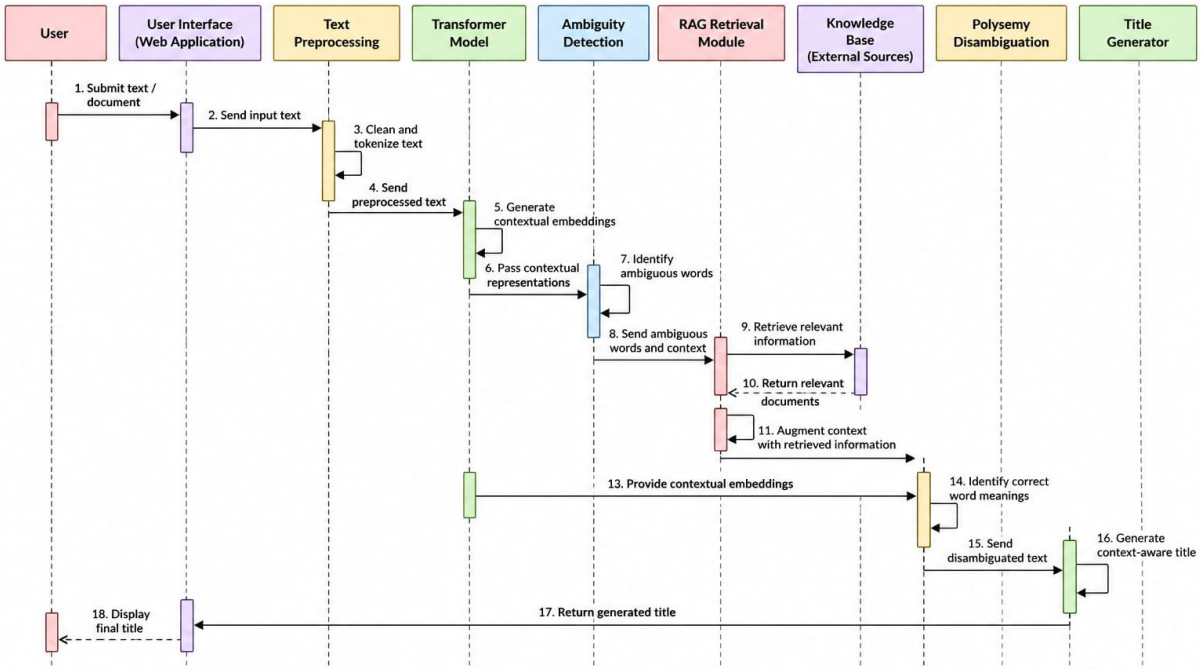


Fig 7.4: Sequence Diagram

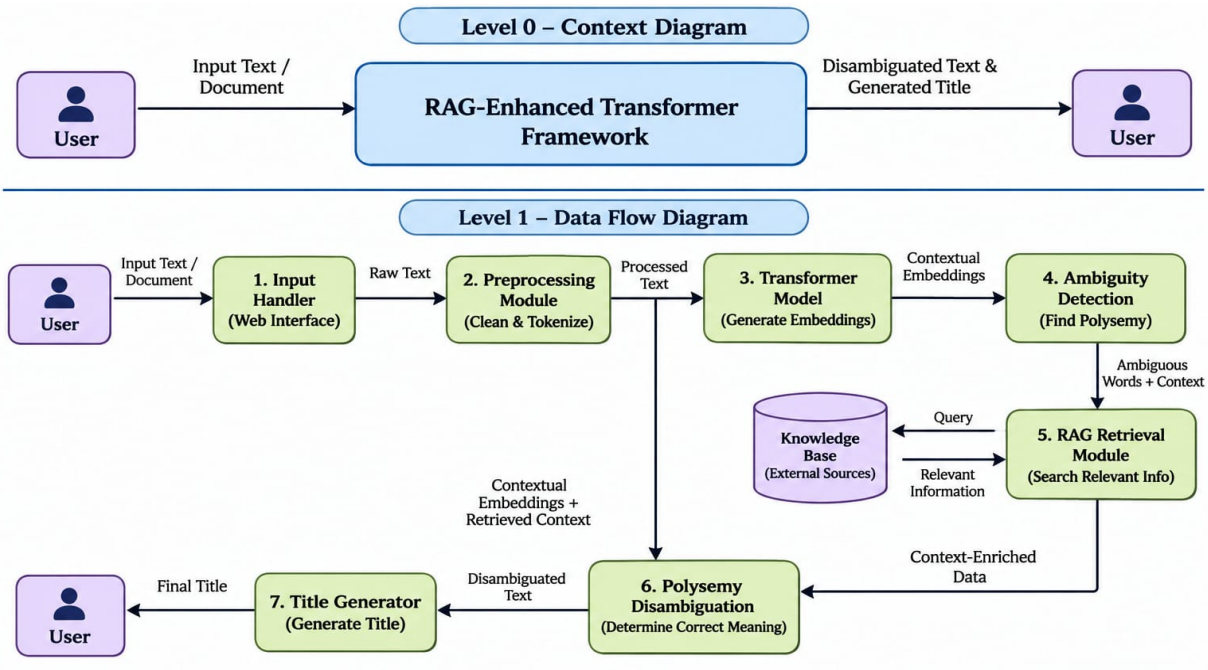


Fig 7.5: Data Flow Diagram

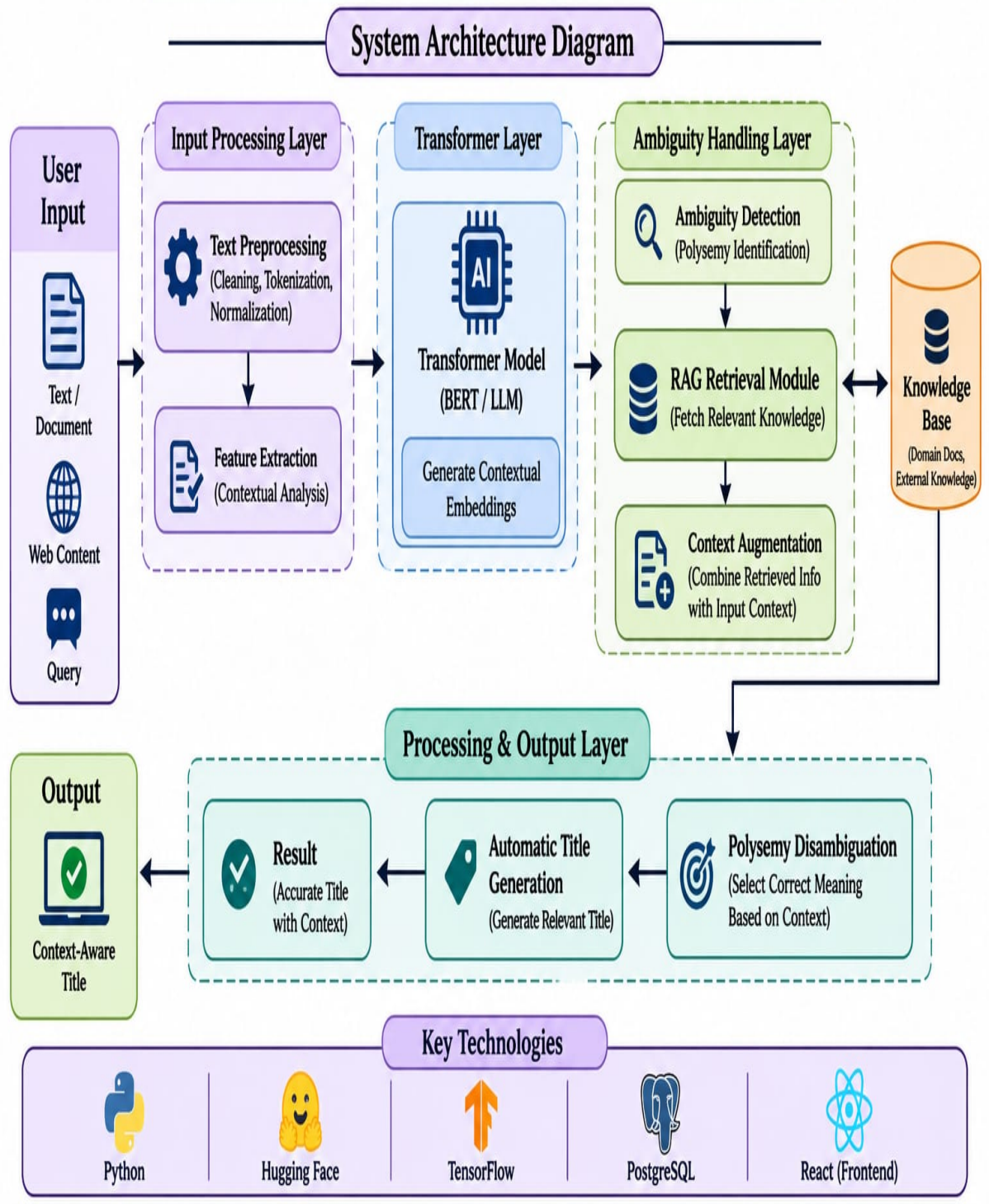


Fig 7.6: System Architecture Diagram

8. CONCLUSION

This paper addressed the foundational challenges of automated polysemy disambiguation in digital news titles, specifically resolving the high latency, static lexical constraints, and inter-class confusion inherent in earlier retrieval-augmented frameworks. By integrating a dynamic external knowledge repository with byte-pair encoded transformer encoders, the proposed framework eliminated vocabulary fragmentation and mitigated the severe context sparsity typical of short headlines.

The deployment of a multi-task Knowledge-Distilled DistilRoBERTa (RAG-DistilRoBERTa-KD) architecture successfully transferred dark knowledge and soft posterior targets from a high-capacity teacher network, significantly curbing computational overhead and training latency while maintaining top-tier disambiguation accuracy.

Furthermore, the introduction of a multi-model Soft-Voting Ensemble—integrating cross-lingual XLM-RoBERTa representations—effectively neutralized off-diagonal classification errors and resolved subtle semantic ambiguities among closely related categories.

Empirical validations confirmed that this architectural synergy achieves superior precision, recall, and macro F1-scores over conventional baselines, offering a resilient, scalable, and low-latency paradigm well-suited for deployment in real-time media monitoring, automated headline analysis, and large-scale digital journalism pipelines.

Furthermore, the automatic title generation component benefits from the enriched contextual information provided by the disambiguation module, producing titles that are more relevant, concise, and semantically aligned with the source content. The framework demonstrates improved performance in terms of accuracy, contextual relevance, and generation quality when compared to traditional Transformer-based approaches. Its ability to combine knowledge retrieval, context reasoning, and language generation makes it highly suitable for applications such as document summarization, content management, digital libraries, news aggregation, and intelligent information retrieval systems.

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